Self Attention Mechanism
(Embedding Vector Dimension and Softmax Sensitivity)

$$w'_{ij} = x_i^T x_j$$

$[X_i]_{1\times d}$

d ↑

↑ $w'_{ij} = x_i^T x_j$

As Dimension of Embedding "d" Increases the Calculated weight value "w" increases

Scale down "Weights" Values
before applying SoftMax

***Scaling Factor** should be related to Embedding Dimension "d"*

$[X_i]_{1\times d}$

d ↓

SoftMax $w_{ij} = \dfrac{\exp w'_{ij}}{\Sigma_j \exp w'_{ij}}$

High Weight Values kill the gradient, and slow down learning

w'_{ij}	SoftMax
1	0.0321
2	0.0871
3	0.2369
4	0.6439

For small "d"

d ↑

w'_{ij}	SoftMax
5	0.0000
10	0.0000
15	0.0067
20	0.9933

For medium "d"

d ↑↑

w'_{ij}	SoftMax
10	0.0000
20	0.0000
30	0.0000
40	1.0000

For large "d"